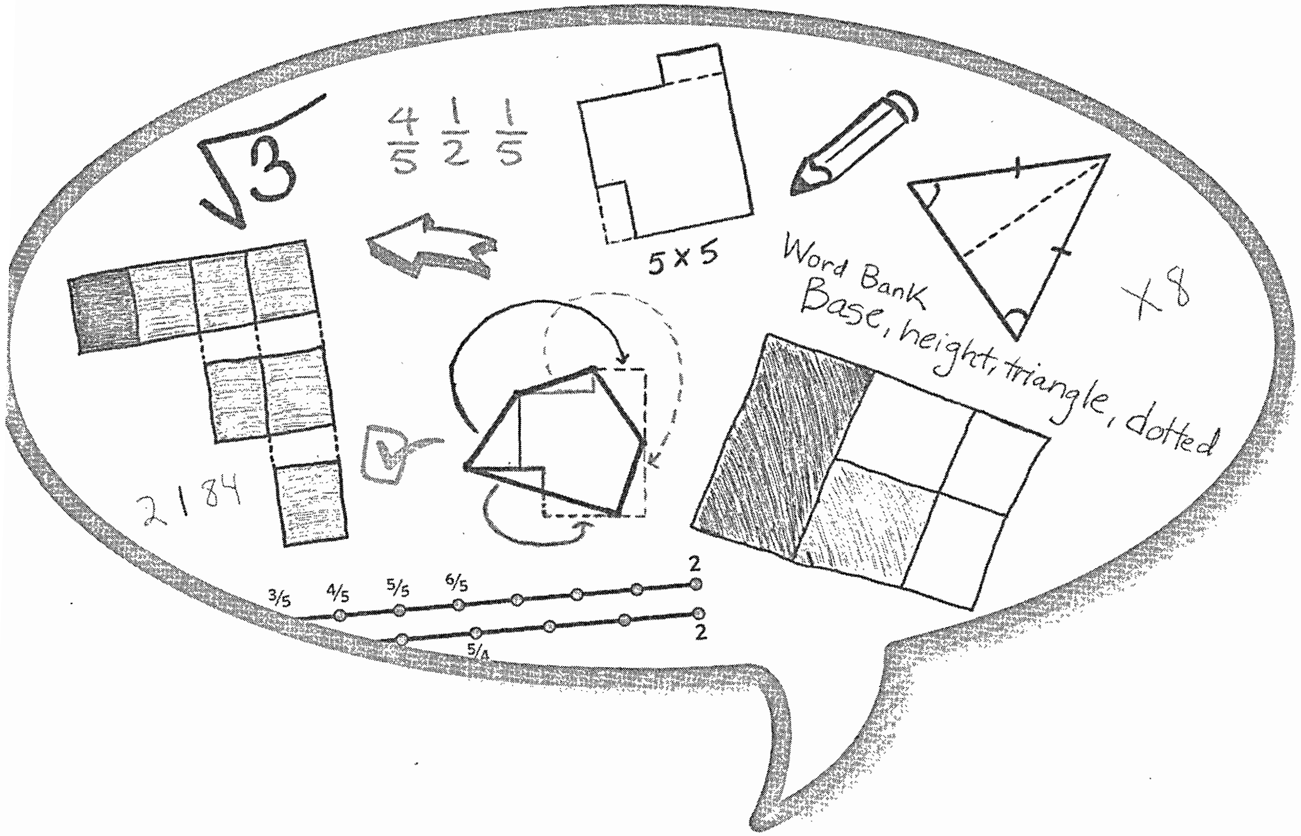

Mathematical Thinking and Communication

Access for English Learners

INCLUDES **ONLINE** PD RESOURCES

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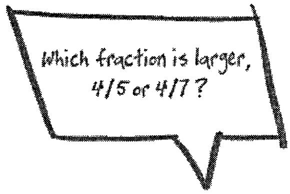
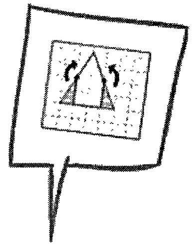




Chapter 5

Designing Tasks to Support Access for English Learners

Mathematics tasks come in a variety of forms, including:

- Tasks meant to prompt recall of facts or prompt practice of learned procedures
- Tasks meant to prompt application of learned procedures in situations that also require some reasoning about the structure of the procedures (for example, think of what is required to find two different sets of six numbers, each set with mean equal to 42)
- Problem-solving/reasoning tasks where paths to solutions are not immediately clear, and where solutions often require more than direct applications of known procedures

Most, if not all, curricula used in schools have a mix of all these. Each type is suited to its purpose, whether that purpose is knowledge recall, skill reinforcement, or application of knowledge and skill in problem solving and reasoning situations. In this book, we have focused on the third category of tasks. That category is richest in opportunities to include visual representations and use of mathematical communication and reasoning, as well as opportunities to collect informative records of student thinking—all for the purpose of providing mathematics access for ELs.

Task Design Principles

While the texture and depth of tasks can shift dramatically, depending on purpose, there are some common task features to concentrate on, which we have distilled into six design principles. We do not insist that every task be true to all six design principles, but we have learned that, over time, we do well to pay heed to all of them. This seems especially true when an overarching purpose is—as it is in this book—to increase access for English learners to mathematical thinking and communication.

1. *The Content Principle.* Does the problem involve concepts or procedures that are recognizably connected to content and mathematical practice standards for the different grades? Will you and your students see the relevance of the content to classroom work?
2. *The Evidence of Thinking Principle.* Will the response require students to think, beyond mere application of procedures, and to reveal their thinking about the task?
3. *The Wide Access Principle.* Will all students, including those who typically struggle in mathematics, find productive as well as engaging ways to enter the problem?
4. *The Multimodal Principle.* Does the problem invite visual, e.g., diagrammatic, approaches? Does it prompt verbal or written explanations? Will opportunities arise for teachers to help students develop and reinforce effective communication and academic language?
5. *The Closure Principle.* Will students (and their teachers) find adequate and appropriate ways to terminate their work on the problem—perhaps not the same closure for all?
6. *The Extension-Generalization Principle.* Will the procedures, explanations, constructions, conjectures, and so on produced by the students apply to other situations?

Attention to these principles can help shape tasks to meet your instructional needs, especially where those needs involve helping ELs become more mathematically proficient. Below are two examples, drawn from two topics that are familiar to all middle school students and their teachers: fractions and area.

Example 1: Fractions

Content Principle

We start by considering the aspects of fractions that make them so important and relevant to student success. For one, proficiency with fractions is an essential prerequisite for success in mathematics topics that are featured in, then extend beyond, middle grades. More specifically, for students to succeed in algebra, they must understand that fractions serve as more than representations of parts of wholes, that they also represent numbers with *magnitude*.[1] What does a task look like that builds knowledge and skills in using fractions as numbers with magnitude? Think about the kinds of tasks we give younger students to see if they understand magnitude in whole numbers. We might want to know if they can successfully compare two ("Which is larger, 340 or 430?"); in addition, we might want to know if they have a sense of the relative magnitudes ("What is a whole number that is between 340 and 430?"). We can do the same with fractions, so let's take the following as a task starting point:

Which is the larger number, $^4/_7$ *or* $^4/_5$*?*

Find a fraction between $^4/_7$ *and* $^4/_5$.

Evidence of Thinking Principle

By "evidence of thinking," we mean any record students leave and/or make accessible later to themselves, to other learners, and to teachers. Records can benefit the learners themselves because their presence can lighten cognitive load that would build quickly toward overload if the mind had to keep it all internalized. Records can also enlighten other learners because they can be used to piece together a narrative of thinking about a task, as well as give them new ways to think about their own work. For teachers, records of thinking can guide formative assessment and instructional interventions, which can be invaluable to teachers of English learners.

[1] M. Schneider and R. S. Siegler, "Representations of the Magnitudes of Fractions," *Journal of Experimental Psychology: Human Perception and Performance* 36, no. 5 (2010): 1227–38.

In order to gauge student mathematical thinking, we need records of what students *did*, of course, but also records of the thinking behind what they chose to do. While sometimes such records can be spoken, usually it is most convenient to seek written records of thinking. In the end, there should be a variety of thinking apparent across students. Otherwise, if student responses look alike, then the task may not be challenging enough, or, in some cases, the teacher's prompts were too leading. Variety in students' thinking is valuable.

Some students might think, "$^4/_7$ is smaller than $^4/_5$ because $^1/_7$ is smaller than $^1/_5$ and so 4 of the $^1/_7$s is smaller than 4 of the $^1/_5$s. To find a fraction between them, take 6, a number between 5 and 7, and so $^4/_6$ is between $^4/_7$ and $^4/_5$." This method applies directly to similar pairs like $^5/_6$ and $^5/_8$, but for $^8/_9$ and $^8/_{10}$, there is no whole number between 9 and 10. Many students in middle grades will try $^8/_{9.5}$ and see this is equivalent to $^{16}/_{19}$, which is a fraction between $^8/_9$ and $^8/_{10}$. So, the method continues to work for them, and they have found "regularity in repeated reasoning," which is the core of Mathematical Practice 8, *Look for and express regularity in repeated reasoning.*

Other students might approach the problem visually, for example, making use of the geometric structure of the number line, consistent with Mathematical Practice 7, *Look for and make use of structure.*

To capture all such examples of student thinking, in student written narrative, we might prompt the records as follows:

Which is the larger number, $^4/_7$ or $^4/_5$? Explain your thinking.

Find a fraction between $^4/_7$ and $^4/_5$. Explain your thinking.

Wide Access Principle

For all students, but especially English learners, a fundamental consideration for access to a task is the wording of the task. Are there too many words? Are there words that may be unfamiliar, particularly to students whose first language is not English? These considerations relate to access to understanding the task. On brief examination, it appears that our wording is brief enough and familiar enough. However, another aspect of access relates to how well students are enabled to proceed with what we have prompted them to do. In our case, we have prompted written explanation. For ELs with low English proficiency, this may be difficult. So, we can incorporate sentence frames:

Which is the larger number, $^4/_7$ or $^4/_5$? Explain your thinking: I know that ___ is a larger number than ___ because ______.

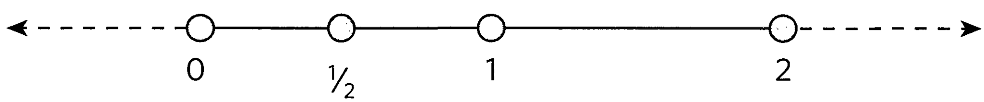
Find a fraction between $^4/_7$ and $^4/_5$. Explain your thinking: The number ___ is between $^4/_7$ and $^4/_5$ because_____.

We try, in offering tasks for ELs, to balance the prompts for verbal explanations with opportunities, if not direct prompts, to use visual representations of their thinking, which we take up next.

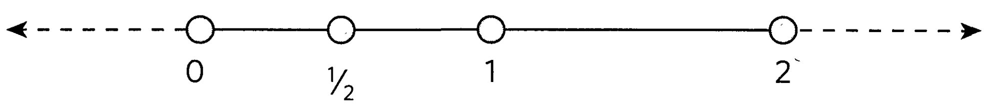
Multimodal Principle

We have already invoked the multimodal principle in asking for students to write explanations. With English learners, it often is important to expand the modes further by inviting or requiring the use of visual representations of thinking. So, we might have:

Which is the larger number, $^4/_7$ or $^4/_5$? Show your thinking on the number line. Then explain your thinking by completing this sentence: My number line shows that ___ is a larger number than ___ because ______.



Find a fraction between $^4/_7$ and $^4/_5$. Show your thinking on the number line. Explain your thinking by completing this sentence: My number line shows that ___ is between $^4/_7$ and $^4/_5$ because _____.



We also have found it valuable to have students work in pairs on tasks, and we incorporate a prompt to "explain to your partner what you did" before they write explanations. This expands the number of modes even further—combining oral explanation with written explanation and visual representation.

Levels of Closure Principle

For some more open-ended tasks, recognizing when one is at a point of closure can be challenging. (See Example 2, below, for a case.) However, in this fraction task, the closure boundaries are pretty clear. Some students respond to the first part, describing their thinking on comparing the two fractions,